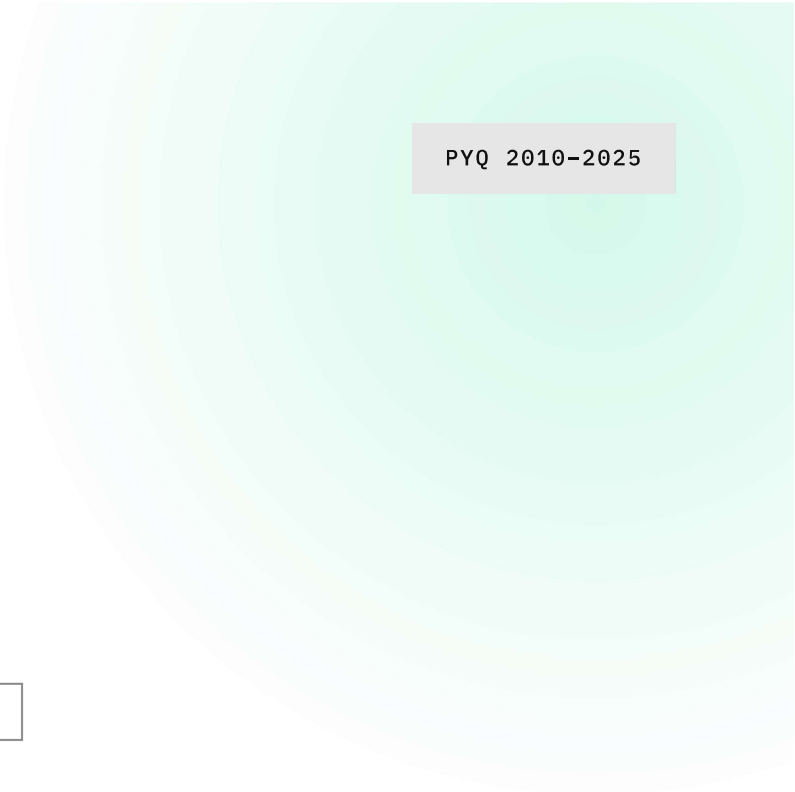




PYQ 2010–2025

WBUHS BIOCHEMISTRY • MODULE 02

Carbohydrate Metabolism

Glycolysis • TCA Cycle • HMP Shunt • Glycogen • Gluconeogenesis

SUBJECT

Biochemistry — Paper 1

EXAM RELEVANCE

Group A • B • C • D

PYQ FREQUENCY

12+ times (2010–2025)



★ APPEARED IN WBUHS — YEARS

2010

2010-S

2012-S

2013

2014-S

2015-S

2016

2016-S

2017

2017-S

2018-S

2019

2019-S

2021

2022

2023

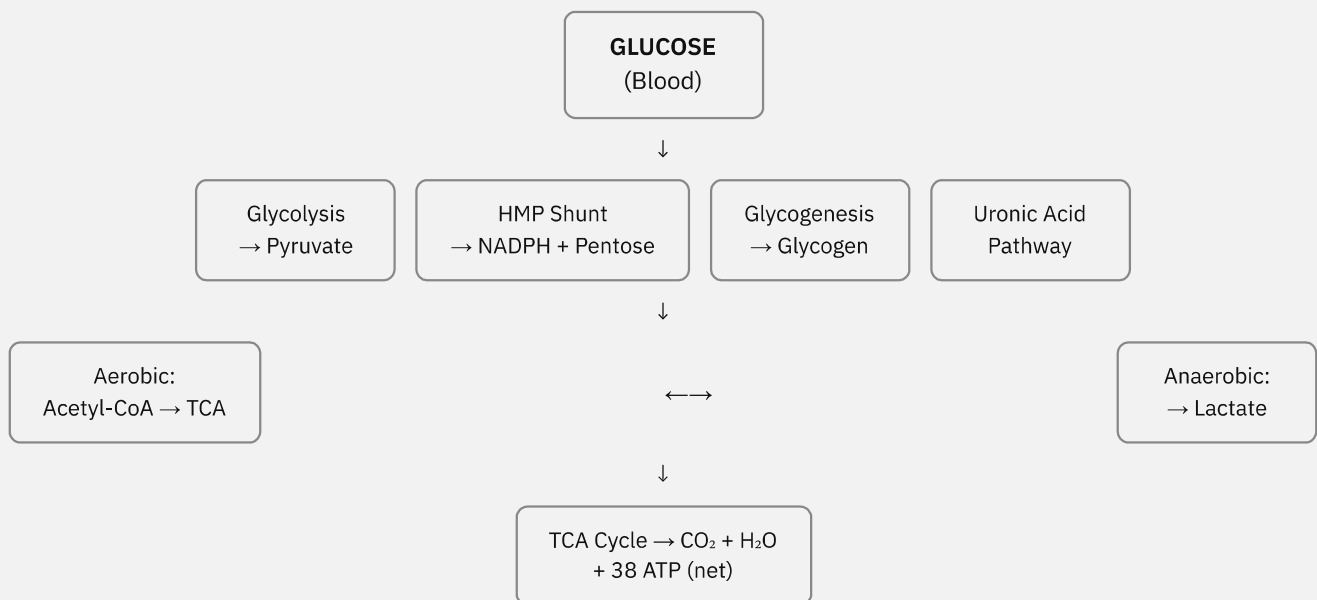
2025

01 The Big Picture

🩺 WHY THIS MATTERS CLINICALLY

Carbohydrate metabolism is the **central hub of energy production** in the body. Every pathway here connects to a clinical disease — Diabetes (glycolysis regulation), G6PD deficiency (HMP shunt), Von Gierke's disease (glycogen storage), Lactic acidosis (pyruvate dehydrogenase deficiency). Your examiner will test this as Group A almost every year because it demands flowcharts, enzyme knowledge, and clinical correlation all in one answer.

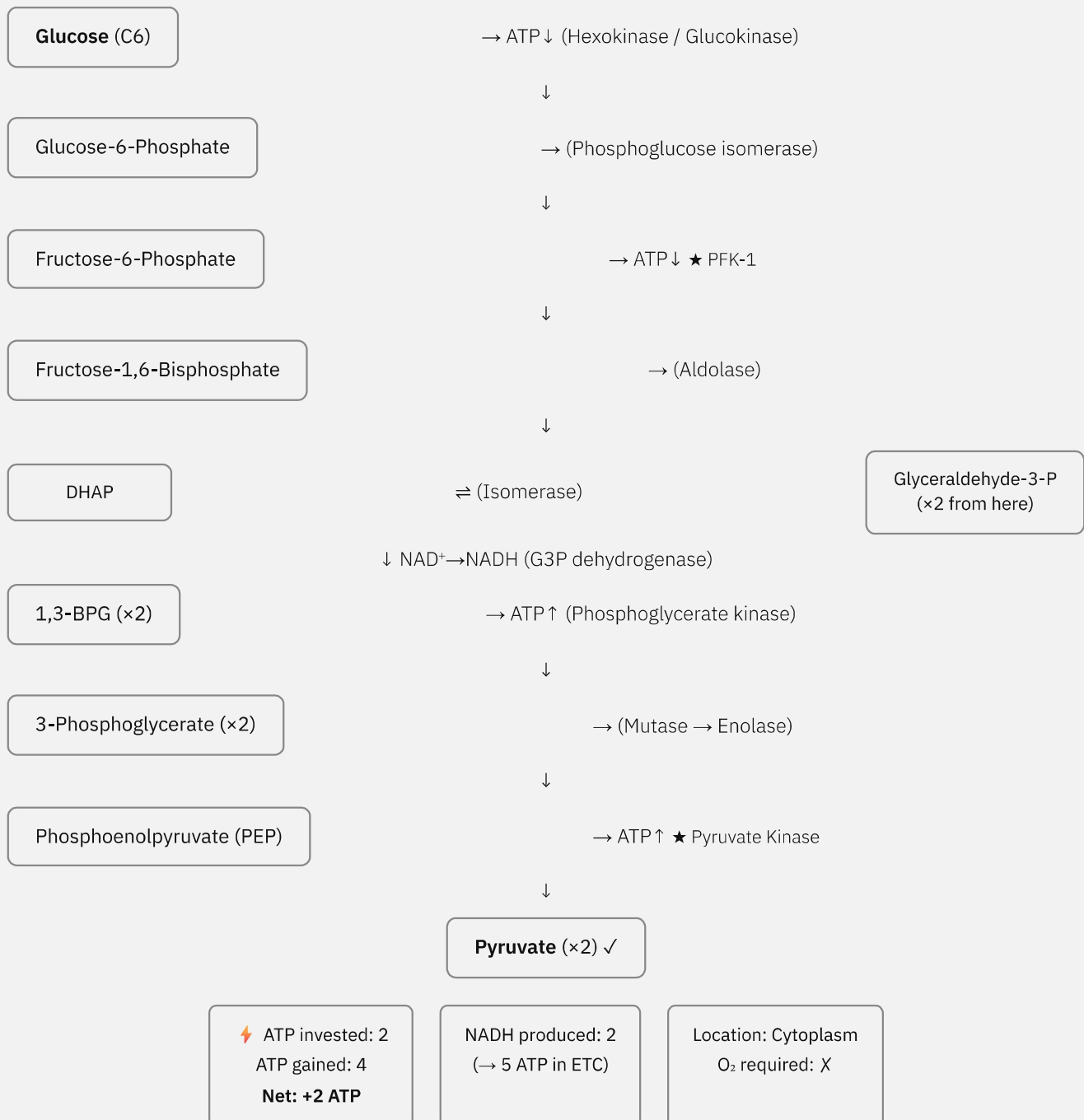
// MASTER OVERVIEW — FATE OF GLUCOSE



📌 ONE-LINER DEFINITION

Glycolysis is the **cytoplasmic, oxygen-independent** breakdown of one molecule of glucose (6C) into **two molecules of pyruvate (3C)**, yielding a net gain of **2 ATP + 2 NADH**.

// GLYCOLYSIS – 10 STEPS FLOWCHART (EXAM-READY)



**Rate-Limiting Step:**

RATE LIMITING STEP

Phosphofructokinase-1 (PFK-1) — catalyses fructose-6-P → fructose-1,6-bisphosphate. Called the "**pacemaker of glycolysis**." Activated by AMP, ADP, fructose-2,6-bisphosphate. Inhibited by ATP, citrate, low pH.



Anaerobic Glycolysis (in RBCs, muscle during sprint): Pyruvate → Lactate (by LDH). This regenerates NAD^+ so glycolysis can continue without oxygen. **RBCs have no mitochondria** — 100% dependent on anaerobic glycolysis for ATP.



Rapoport-Luebering Shunt (RBC specific): 1,3-BPG → **2,3-BPG** (bypasses substrate-level phosphorylation → no ATP made here). 2,3-BPG stabilises T-form of Hb → ↓ O_2 affinity → promotes O_2 delivery to tissues.



CLINICAL LAB CORRELATION: FLUORIDE VIALS

- ▶ Blood for glucose estimation is collected in fluoride vials because Fluoride irreversibly inhibits the enzyme Enolase, stopping RBC glycolysis during transport.

IRREVERSIBLE STEPS OF GLYCOLYSIS (BYPASS POINTS FOR GLUCONEOGENESIS)

Step 1 Glucose → G6P — by **Hexokinase/Glucokinase** (bypassed in GNG by Glucose-6-phosphatase)

Step 3 F6P → F-1,6-BP — by **PFK-1** (bypassed in GNG by Fructose-1,6-bisphosphatase)

Step 10 PEP → Pyruvate — by **Pyruvate Kinase** (bypassed in GNG by Pyruvate carboxylase + PEPCK)

03 Pyruvate Dehydrogenase Complex

// PYRUVATE → ACETYL-COA (BRIDGE REACTION)

Pyruvate (3C)

→ CO₂↑ + NAD⁺→NADH

Acetyl-CoA (2C)

PDH COMPLEX — 3 ENZYMES + 5 COENZYMES

E1 — Pyruvate decarboxylase

Coenzyme: **TPP** (Thiamine B1)

E2 — Dihydrolipoyl transacetylase

Coenzyme: **Lipoic acid + CoA**

E3 — Dihydrolipoyl dehydrogenase

Coenzyme: **FAD + NAD⁺**



PDH regulation: Activated by: ADP, CoA, NAD⁺, Ca²⁺, pyruvate. Inhibited by: ATP, Acetyl-CoA, NADH (products inhibit). PDH kinase phosphorylates (inactivates) PDH; PDH phosphatase dephosphorylates (activates) it.



Clinical: Thiamine (B1) deficiency → PDH complex failure → pyruvate accumulates → converted to lactate → **Lactic acidosis**. Presents as Wernicke's encephalopathy in alcoholics. Transketolase (HMP shunt) also needs TPP — used as diagnostic marker.

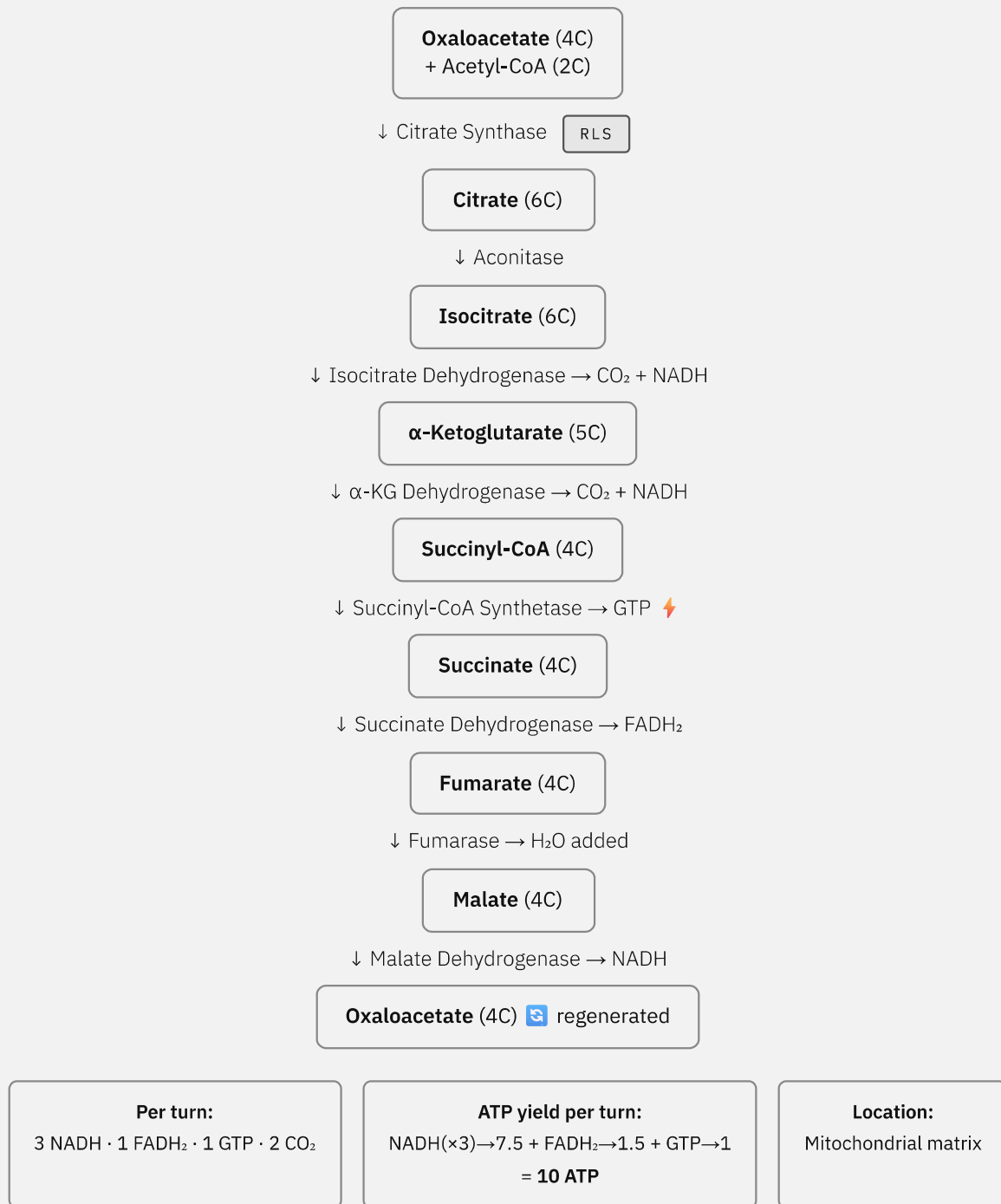
04 TCA / Krebs Cycle

★ EXAMINER FAV

📌 ONE-LINER DEFINITION

The TCA cycle is an **amphibolic** (both anabolic + catabolic), **mitochondrial** cyclic pathway where 1 Acetyl-CoA (2C) condenses with Oxaloacetate (4C) to generate **3 NADH + 1 FADH₂ + 1 GTP + 2 CO₂** per turn.

// TCA CYCLE — COMPLETE FLOWCHART



IRREVERSIBLE (REGULATED) STEPS OF TCA

1	Citrate Synthase	— Oxaloacetate + Acetyl-CoA → Citrate	RLS	— Inhibited by: ATP, NADH, Succinyl-CoA, Citrate
3	Isocitrate Dehydrogenase	— Activated by: ADP, Ca ²⁺ . Inhibited by: ATP, NADH		
4	α-Ketoglutarate Dehydrogenase	— Similar regulation to PDH complex. Needs TPP, Lipoic acid, CoA, FAD, NAD ⁺		



TCA is AMPHIBOLIC — know why this is asked: Catabolic (breaks down Acetyl-CoA $\rightarrow$ CO₂ + energy). Anabolic (intermediates drawn off for biosynthesis): α -KG $\rightarrow$ glutamate $\rightarrow$ amino acids; Succinyl-CoA $\rightarrow$ porphyrins/haem; OAA $\rightarrow$ gluconeogenesis.



TCA Cycle Inhibitors (Frequent PYQ): Fluoroacetate (rat poison) inhibits Aconitase. Malonate is a classic competitive inhibitor of Succinate Dehydrogenase. Arsenite inhibits α -Ketoglutarate dehydrogenase.

05 HMP Shunt (Pentose Phosphate Pathway)

// HMP SHUNT — OXIDATIVE PHASE (HIGH YIELD)

Glucose-6-Phosphate

↓ G6PDH RLS $\rightarrow$ NADPH + CO₂

6-Phosphogluconolactone

↓ $\rightarrow$ NADPH

Ribulose-5-Phosphate (5C)

↓ Non-oxidative phase (Transketolase + Transaldolase)

Ribose-5-P
(nucleotides)

F-6-P + G-3-P
(re-enters glycolysis)

Products:

2 NADPH · Ribose-5-P · CO₂

Location:

Cytoplasm

Active in: Liver, RBCs,
Adrenal cortex, Testis



Rate-Limiting Step: RATE LIMITING STEP **G6PD (Glucose-6-Phosphate Dehydrogenase)** — generates NADPH. Activated by NADP⁺. Inhibited by NADPH.



Functions of NADPH from HMP shunt: (1) Reduces glutathione (GSH) — protects RBCs from oxidative damage. (2) Required for fatty acid synthesis. (3) Required for cholesterol synthesis. (4) Cytochrome P450 hydroxylation. (5) Phagocyte respiratory burst.

G6PD DEFICIENCY — CLASSIC PYQ CLINICAL CASE

- ▶ **X-linked recessive** — males more affected
- ▶ G6PD deficiency → ↓ NADPH → ↓ reduced glutathione (GSH) → RBCs cannot neutralise oxidative stress
- ▶ Trigger: Primaquine (antimalarial), Dapsone, fava beans, infection → **Heinz bodies** form → RBC membrane damage → **Haemolytic anaemia**
- ▶ Lab: ↓ G6PD activity, Heinz bodies on peripheral smear, ↑ bilirubin (indirect), haemoglobinuria (dark urine)
- ▶ **Protection against Malaria** — G6PD deficiency common in malaria-endemic regions

06

Glycogen Metabolism

★ EXAMINER FAV

// GLYCOGENESIS — SYNTHESIS FLOWCHART

Glycogenin: core protein primer that auto-catalyses attachment of the first few glucose molecules.

↓

Glucose → G6P → G1P

→ UTP

UDP-Glucose
(activated form)

↓ Glycogen Synthase **RLS** (adds α-1,4 links)

Linear Glycogen Chain

→ Branching Enzyme

Branched Glycogen
(α-1,6 branch points)

// GLYCOGENOLYSIS - BREAKDOWN FLOWCHART

Glycogen

→ Phosphorylase **RLS**
(cleaves α -1,4 links + Pi)

G1P → G6P

↓ (at branch points)

Debranching enzyme
(α -1,6 glucosidase)

→ Free glucose released

In LIVER:

G6P → Glucose (by G6Pase)
→ released into blood
(maintains blood glucose)

In MUSCLE:

G6P → Glycolysis directly
(no G6Pase → can't
release free glucose)

// HORMONAL REGULATION VIA CAMP (GLUCAGON/EPINEPHRINE)

Glucagon / Epinephrine

→ GPCR → Adenylyl cyclase

↓

ATP → cAMP **2ND MESSENGER**

→ Protein Kinase A (PKA)

↓ PKA phosphorylates:

Glycogen Synthase
→ INACTIVE X

Phosphorylase Kinase
→ ACTIVE ✓

↓ Glycogenesis

↑ Glycogenolysis

↓

Net result: Glucose released → Blood glucose ↑

Insulin has OPPOSITE effect → activates Protein Phosphatase → activates Glycogen Synthase + inactivates Phosphorylase → promotes Glycogenesis

- ▶ **Type I — Von Gierke's:** G6Pase deficiency → ↑ Glycogen in liver/kidney → Fasting hypoglycaemia, hepatomegaly, **Hyperuricaemia** (classic PYQ trap)
- ▶ **Type II — Pompe's:** Acid maltase (α -1,4 glucosidase in lysosome) deficiency → Cardiomegaly, hypotonia, early death
- ▶ **Type III — Cori's:** Debranching enzyme deficiency → Abnormal glycogen with short outer chains
- ▶ **Type V — McArdle's:** Muscle phosphorylase deficiency → Exercise intolerance, cramps, no lactate rise after exercise
- ▶ **Type VI — Hers':** Liver phosphorylase deficiency → Mild hypoglycaemia

07 Gluconeogenesis

 DEFINITION

Synthesis of **new glucose from non-carbohydrate precursors**. Occurs mainly in **liver (90%)** and kidney cortex (10%) during fasting/starvation. Not simply the reverse of glycolysis — three irreversible steps of glycolysis are bypassed by unique enzymes.



Precursors (GALA-P mnemonic): Glycerol (from TAG breakdown) · Alanine + other amino acids (glucogenic AAs) · Lactate (Cori cycle) · Acetyl-CoA (indirectly, via OAA) · Propionate (odd-chain FA → Succinyl-CoA)

// 3 BYPASS STEPS OF GLUCONEOGENESIS

Glycolysis:
Pyruvate → PEP
(Pyruvate Kinase)

→ Bypassed by →

Pyruvate → OAA
(Pyruvate Carboxylase, mito) B7 BIOTIN
OAA → PEP
(PEPCK, cytoplasm)

Glycolysis:
F6P → F-1,6-BP
(PFK-1)

→ Bypassed by →

F-1,6-BP → F6P
(Fructose-1,6-Bisphosphatase)

Glycolysis:
G6P → Glucose
(Hexokinase)

→ Bypassed by →

G6P → Glucose
(Glucose-6-Phosphatase)
— only in LIVER/KIDNEY



Cori Cycle (Lactic acid cycle): Muscle lactate → blood → liver → gluconeogenesis → glucose → blood → back to muscle. Net: transfers metabolic burden from muscle to liver. Energy-expensive: It costs exactly 6 ATP equivalents (4 ATP + 2 GTP) in the liver to synthesize 1 molecule of glucose from 2 pyruvates, compared to the 2 ATP gained in muscle glycolysis.



Regulation by Fructose-2,6-Bisphosphate (F-2,6-BP): CLASSIC PYQ 2017 F-2,6-BP activates PFK-1 (glycolysis) and inhibits F-1,6-BPase (GNG) — a single molecule that switches between the two pathways. Insulin ↑ F-2,6-BP → glycolysis. Glucagon ↓ F-2,6-BP → GNG.

// FED VS FASTING STATE

FED STATE (↑ Insulin)

- Glucose uptake ↑ (GLUT4 in muscle/fat)
- Glycolysis ↑
- Glycogenesis ↑
- Lipogenesis ↑
- Gluconeogenesis ↓
- Protein synthesis ↑

FASTING STATE (↑ Glucagon)

- Glycogenolysis ↑ (liver first)
 - Gluconeogenesis ↑ (after 4–6h)
 - Lipolysis ↑ → FFA + Ketones
 - Protein catabolism ↑ (AA for GNG)
 - Glycolysis ↓
 - Glycogenesis ↓
-

09 Fructose & Galactose Metabolism

// FRUCTOSE METABOLISM — NORMAL & ABNORMAL

Fructose

→ Fructokinase (liver) → Fructose-1-P

↓ Aldolase B → DHAP + Glyceraldehyde

Enters Glycolysis ✓

Essential Fructosuria: Fructokinase deficiency → fructose in urine. Benign.

Hereditary Fructose Intolerance (HFI): Aldolase B deficiency → Fructose-1-P accumulates → **Hypoglycaemia + liver damage.** Avoid sucrose/fructose.

// GALACTOSE METABOLISM — NORMAL & ABNORMAL

Galactose

→ Galactokinase → Galactose-1-P

↓ Galactose-1-P Uridyl Transferase (GALT) → UDP-Galactose → UDP-Glucose → G1P

Enters Glycolysis ✓

Galactosaemia (GALT deficiency): Galactose-1-P accumulates → liver damage, cataracts (galactitol in lens), intellectual disability. **Newborn screening mandatory.**

Galactokinase deficiency: Milder — only cataracts (galactitol accumulates).

10 Memory Mnemonics

🧠 MNEMONIC 1 — TCA INTERMEDIATES IN ORDER

"Citrate Is Krebs' Starting Substrate For Making Oxaloacetate"

Citrate → Isocitrate → α-Ketoglutarate → Succinyl-CoA → Succinate → Fumarate → Malate → Oxaloacetate

🧠 MNEMONIC 2 — IRREVERSIBLE GLYCOLYSIS STEPS

"Hungry People Feel Pale" → HK, PFK-1, PK

Hexokinase (step 1) · Phosphofructokinase-1 (step 3) · Pyruvate Kinase (step 10) — these 3 are irreversible and are bypassed in gluconeogenesis.

🧠 MNEMONIC 3 — PDH COMPLEX COENZYMES

"The Lovely Cat Finds Nice Accommodation"

TPP (B1) · Lipoic acid · CoA (B5 pantothenic acid) · FAD (B2) · **NAD⁺** (B3) · Note: All B-vitamins!

🧠 MNEMONIC 4 — GLYCOGEN STORAGE DISEASES

"Very Poor Carbohydrate Metabolism" → Types I, II, III, V

Type I = **V**on Gierke (G6Pase) · Type II = **P**ompe (Acid maltase) · Type III = **C**ori (Debranching) · Type V = **M**cArdle (Muscle phosphorylase)

🧠 MNEMONIC 5 — GNG PRECURSORS

"GALA Party" → Glycerol, Amino acids, Lactate, Alanine / Propionate

All glucogenic amino acids → glucose. Fatty acids (even-chain) CANNOT → glucose (no net conversion from Acetyl-CoA). Odd-chain FA → Propionyl-CoA → Succinyl-CoA → CAN enter GNG.



HIGH-YIELD CLINICAL LINKS

- ▶ **Diabetes Mellitus:** Insulin deficiency → ↓ GLUT4 → ↓ glucose uptake → Hyperglycaemia. ↑ Glycogenolysis, ↑ GNG, ↑ Lipolysis, ↑ Ketogenesis → Diabetic Ketoacidosis (DKA). HbA1c reflects average glucose over 3 months (glycated Hb).
 - ▶ **Lactic Acidosis:** Anaerobic glycolysis excess (shock, sepsis) or PDH deficiency → Lactate accumulates → ↓ pH. High Anion Gap Metabolic Acidosis.
 - ▶ **Von Gierke's + Hyperuricaemia:** ↑ G6P → ↑ HMP shunt → ↑ Ribose-5-P → ↑ Purine synthesis → ↑ Uric acid. Classic PYQ trap — explain the link!
 - ▶ **Galactosaemia → Cataract:** Galactose → Galactitol (by aldose reductase) → Accumulates in lens (avascular) → Osmotic damage → Cataract. Treat: eliminate galactose/lactose from diet.
 - ▶ **Uncontrolled DM → Cataract:** Glucose → Sorbitol (aldose reductase) → Accumulates in lens → Same mechanism as galactosaemia.
 - ▶ **Primaquine + G6PD deficiency:** Oxidative drug → ↑ ROS → ↓ NADPH (G6PD absent) → ↓ GSH → Heinz bodies → Haemolysis. Tested in clinical MCQs.
-

Q1 Why can't RBCs use the TCA cycle? ↑

RBCs have no mitochondria — TCA cycle and oxidative phosphorylation are mitochondria-dependent. RBCs rely entirely on anaerobic glycolysis for ATP. They also use HMP shunt to generate NADPH for maintaining reduced glutathione (protection against oxidative damage).

Q2 Why does Von Gierke's disease cause hyperuricaemia? ↑

G6Pase deficiency → G6P accumulates → excess G6P shunted into HMP pathway → excess Ribose-5-phosphate → increased de novo purine synthesis → excess purines → increased uric acid production → hyperuricaemia and gout.

Q3 Why is gluconeogenesis NOT the simple reverse of glycolysis? ↑

Three irreversible steps of glycolysis cannot be reversed. These are catalysed by Hexokinase, PFK-1, and Pyruvate Kinase. GNG uses unique bypass enzymes: Glucose-6-phosphatase, Fructose-1,6-bisphosphatase, and the Pyruvate Carboxylase + PEPCK system respectively.

Q4 What is the role of 2,3-BPG? Why is it important in stored blood? ↑

2,3-BPG binds to the β -chains of deoxy-Hb, stabilising the T (tense/deoxy) form, reducing O_2 affinity → promotes O_2 delivery to tissues. In stored blood (bank blood), 2,3-BPG levels fall → Hb has increased O_2 affinity → poor O_2 delivery to recipient's tissues (stored blood paradox).

Q5 Why is TCA called amphibolic? ↑

Amphibolic = both anabolic and catabolic. Catabolic: oxidises Acetyl-CoA → CO_2 + energy. Anabolic: intermediates are precursors for biosynthesis — α -KG → glutamate → amino acids; Succinyl-CoA → haem synthesis; OAA → gluconeogenesis; Citrate → fatty acid synthesis.

Q6 Why does muscle lack glucose-6-phosphatase but liver has it? ↑

Liver's role is to maintain blood glucose homeostasis — it must release free glucose into circulation during fasting. G6Pase allows dephosphorylation of G6P → free glucose → export. Muscle is a glucose consumer — it needs G6P to stay within the cell for glycolysis. If muscle had G6Pase, glucose would leak out rather than be used for energy.

Q7 Fat can be synthesised from glucose but glucose cannot be synthesised from fat — justify. ↑

Glucose → Acetyl-CoA (irreversible via PDH). Fatty acids → β -oxidation → Acetyl-CoA. Acetyl-CoA cannot be converted back to pyruvate/OAA in net amounts in mammals — the PDH reaction is irreversible. Acetyl-CoA can only enter TCA (catabolic) or be used for lipid synthesis, not for GNG. Exception: glycerol backbone of TAG can yield glucose (via DHAP).

Q8 What is the significance of oral glucose in ORS (Oral Rehydration Solution)? ↑

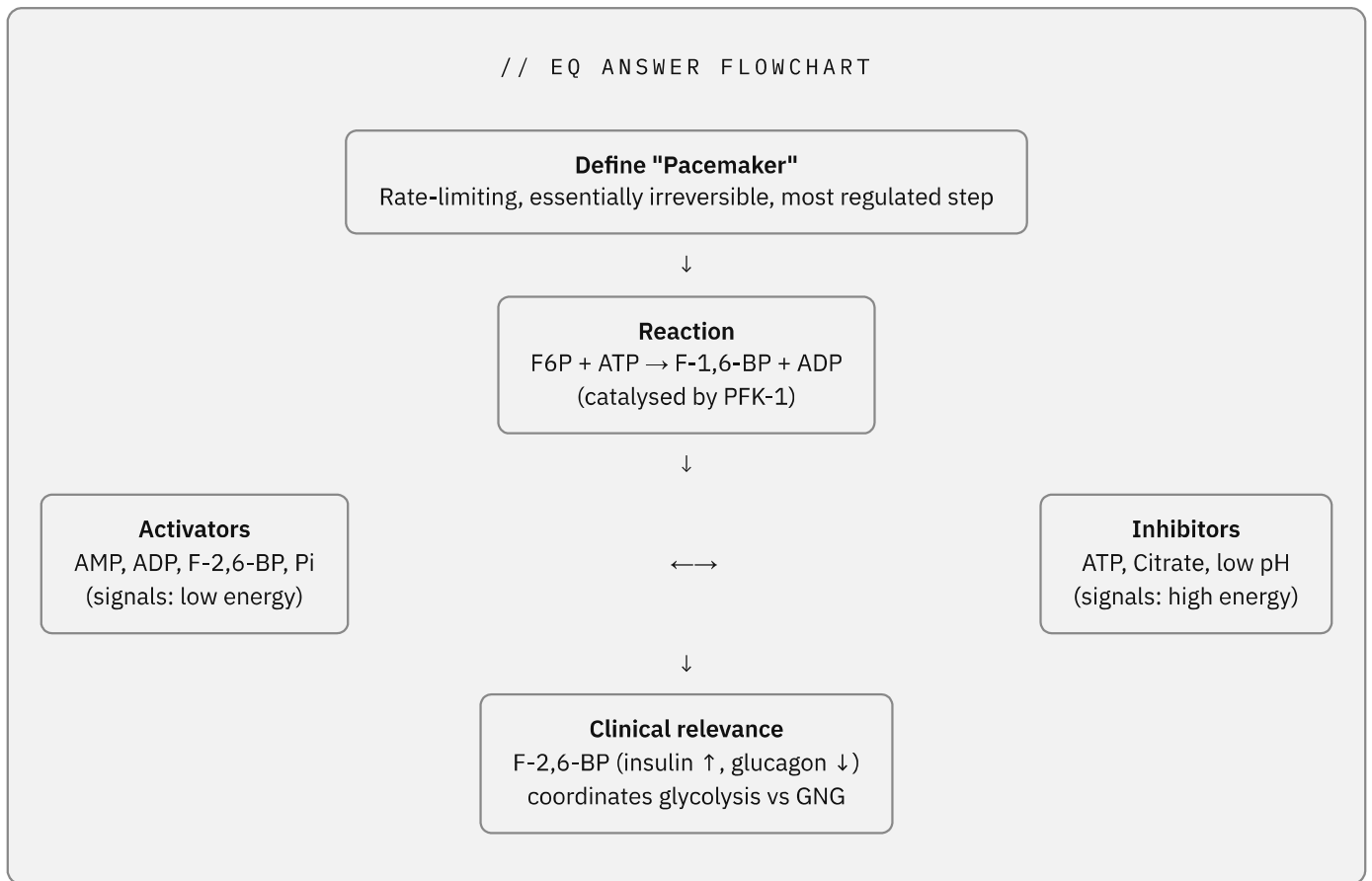
Glucose in ORS co-transporters sodium via SGLT1 (Na^+ -glucose co-transporter) in the intestinal mucosa — a secondary active transport system. This Na^+ uptake drags water with it by osmosis, rehydrating the patient. Glucose doesn't just provide energy — its primary role in ORS is to drive Na^+ absorption even in secretory diarrhoea where other Na^+ transporters are inhibited.

13 Group-D EQ Template

 HOW TO ANSWER: "PHOSPHOFRUCTOKINASE-I IS THE PACEMAKER OF GLYCOLYSIS"

Use your teacher's flowchart formula — Define → Mechanism Flowchart → Kinetics/Regulation → Clinical consequence.

// EQ ANSWER FLOWCHART



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